

# Konvert

Your Smart Sales Engine





Your Smart Sales Engine

To design and develop an AI-integrated business management and sales tracking ecosystem that optimizes multi-tier enterprise hierarchies, streamlines distribution workflows, and provides a localized, secure alternative for data-driven decision making.

**Undertaken By:**

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# Dedication

**To our parents,**

the silent engineers of our souls, whose loyal belief and quiet offerings handed the bedrock upon which our bourns were erected. You gave us the roots to stay predicated and the bodies to seek the horizon.

**To our instructors,**

especially Mr. Hafiz Ahsan Arshad, who tutored us that the language of law is but a tool to rewrite the world, and that every specialized challenge is an assignation to introduce.

**And to the land that nurtured us.**

May this endeavor stand as a testament to the brilliance of our own soil, a small donation toward a future where our technology is as flexible and independent as our spirit.

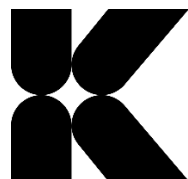
# Acknowledgement

We express our profound gratefulness to Almighty Allah for granting us the strength, health, and perseverance needed to take over and complete this ambitious bid. We're deeply obliged to the University of Management and Technology (UMT) and the Department of Computer Science for furnishing the academic foundation and coffers necessary to transfigure a complex vision into a functional reality.

Our sincerest gratefulness goes to our advisor, Mr. Hafiz Ahsan Arshad, for his specialized mentorship and unwavering support. His guidance was necessary in helping us navigate the complications of enterprise- position software development and AI integration.

A special and significant note of appreciation is extended to the CEO of Bristol. We're truly recognized by the direct support and trust he placed in us to break a critical, real- world business problem for his association. His amenability to give us with direct access to his company's functional perceptivity was the foundation of this design. We also admit the team at Almashreqs for their cooperation in easing the integration between the mobile ecosystem and the being web structure, icing data thickness across the platform.

Eventually, we fete the cooperative spirit of every group member — Asim, Obaid, and Farheen. This design is a testament to our collaborative fidelity to creating a localized result that not only fulfills our degree conditions but also serves as a point of pride for our country's technological geography.



## **Abstract**

In Pakistan's commercial landscape, businesses are often held back by fragmented data and complex foreign tools. Konvert is an AI-driven management ecosystem designed to serve as the automated backbone of an enterprise's sales operations. Built for real-world distribution challenges, it optimizes a massive eight-tier hierarchy—from Sales Representatives on the route to the CEO—providing a unified "single source of truth."

By combining AI performance forecasting with real-time route tracking, Konvert automates repetitive sales tasks and reduces management overload under one secure environment. Architected for total scalability, the system specifically aims to boost Return on Investment (ROI) for businesses ranging from SMEs to large enterprises. This project is a commitment to technological sovereignty—replacing flawed foreign legacy software with a high-performance Pakistani engine designed to lead the future of national business management.

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# Introduction

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## 1.1

### Motivation

Entrepreneurs with world-class products often face an immediate digital barrier because the sales management tools currently available are either prohibitively expensive foreign platforms or fragmented legacy systems that are too complex to master quickly. This creates a restrictive environment where businesses lose critical momentum navigating technical friction instead of focusing on their core commercial goals.

Furthermore, a reliance on external software introduces serious security risks and a total loss of data sovereignty. In the high-stakes field of sales management, the absence of a localized, secure, and intuitive engine means that vital business intelligence is often stored on foreign servers or lost in manual paperwork. There is an urgent need for a streamlined solution that allows Pakistani businesses to secure their data and optimize their sales workflows without the burden of unnecessary complexity or cost.

## 1.2

### Project Overview

Konvert is a unified, AI-integrated sales ecosystem designed to synchronize field operations with strategic oversight, serving as the definitive automated backbone for Pakistani enterprises.

Moving beyond the limitations of traditional, reactive tracking software, Konvert establishes a high-performance technical bridge between mobile field operations and centralized web infrastructures. The platform is specifically architected to manage and optimize a complex eight-tier organizational hierarchy, ensuring that data generated by Sales Representatives on the route is instantly synchronized with executive dashboards. By integrating directly with existing web management systems, such as the Almashreqs infrastructure for Bristol, Konvert



eliminates the fragmentation and data silos that lead to operational blind spots, establishing a real-time “Single Source of Truth” for the entire organization.

The core of the system is a proactive intelligence engine that utilizes machine learning and advanced APIs to drive ROI through live route tracking, automated tour planning, and AI-driven performance forecasting. These predictive capabilities allow businesses to transition from defensive error-correction to proactive growth, accurately identifying future scalability and market trends. Designed for total enterprise scalability from burgeoning startups to massive distributors, the system automates mundane sales tasks and reduces operational costs while safeguarding national data sovereignty, standing as a secure domestic engine designed to empower the Pakistani workforce and catalyze national economic independence.

### 1.3 **Problem Statement**

Pakistani enterprises are currently paralyzed by a digital deadlock, caught between obsolete regional systems that fail to function and global software giants that are impossible to afford.

The reliance on fragmented, foreign legacy software has created a critical deficit in data integrity, security, and operational oversight. Organizations like Bristol are forced to navigate a dual-threat landscape:

#### **On one side,**

regional platforms like SanSFE offer a broken user experience and a total lack of synchronization with central systems (like Almashreqs), leading to massive data loss

#### **On the other hand,**

world-class solutions like Salesforce or HubSpot remain financially out of reach and operationally over-engineered for the local market.

This Informational asymmetry and a total lack of bi-directional synchronization between field apps and central databases result in reporting error margins often exceeding 20%. This failure to

align an eight-tier hierarchy leads to significant operational leakage and strategic pivots based on obsolete data, causing the systemic erosion of market share and net profitability.

The critical bottlenecks necessitating an immediate solution include:

## **1. Total Data Desynchronization**

The absence of bi-directional sync between mobile field apps and web databases creates a high-stakes environment of record inconsistencies and redundant double-work.

## **2. Operational Stagnation**

A lack of localized, AI-driven route optimization and tour planning forces field representatives into inefficient workflows, resulting in massive resource wastage.

## **3. Strategic Blindness**

Executive leadership (from Managers to the CEO) is forced to operate without real-time visibility or predictive performance forecasting, leading to reactive rather than proactive management.

## **4. Loss of Sovereignty**

Entrusting national commercial intelligence to foreign servers creates deep-seated data-privacy vulnerabilities and long-term geopolitical risks.

## **5. Economic Exclusion**

Prohibitive foreign subscription costs and steep learning curves ensure that local startups and SMEs remain technologically underdeveloped.

Failure to solve these issues will result in a permanent technological dependency, where Pakistani businesses continue to bleed capital to foreign entities while operating on flawed data. Without a secure, domestic automated backbone, our industries will remain trapped in



strategic blindness, unable to compete in a global economy that demands real-time intelligence and technological sovereignty.

You can read more about the problems of sales management in this digital world and its impact here:

[HubSpot: 2025/2026 Sales Trends Report](#)

[Salesforce: 2026 State of Sales Statistics](#)

[MarketsandMarkets: The ROI Gap Analysis \(via MeetRep\)](#)

## 1.4

### Objectives:

Our objective is to develop the definitive automated backbone for Pakistani businesses, transforming fragmented field data into a synchronized engine for strategic growth and national economic resilience. To achieve the following strategic and technical milestones:

#### 1. Establish Absolute Data Synchronization

To engineer a unified, bi-directional engine that bridges the gap between field-level mobile applications and central web databases, ensuring a "single source of truth" across all organizational tiers.

#### 2. Automate Strategic Backbone Workflows

To dismantle management overload by automating administrative sales tasks, reporting cycles, and inventory tracking, allowing leaders to shift from manual oversight to strategic growth.

|                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>3. Optimize Operational Intelligence via AI</b></p> <p>To deploy AI-driven performance analytics and future forecasting modules that transform raw field data into actionable predictions and scalable roadmaps.</p>                  | <p><b>4. Maximize Business ROI through Localized Efficiency</b></p> <p>To increase the Return on Investment (ROI) for Pakistani enterprises by providing elite-tier route optimization and tour planning features at a fraction of the cost of foreign alternatives.</p> |
| <p><b>5. Safeguard National Data Sovereignty</b></p> <p>To develop a secure, domestic business management environment that keeps critical commercial intelligence within Pakistan's borders, mitigating geopolitical and privacy risks.</p> | <p><b>6. Catalyze Enterprise Scalability</b></p> <p>To build a flexible architecture that empowers businesses of all sizes from burgeoning startups to massive eight-tier distributors—to scale their operations without technical friction.</p>                         |

We aim to revolutionize the way businesses operate in Pakistan by transforming sales management from a manual, high-friction chore into a proactive, data-driven engine. By providing an elite-tier technical infrastructure that is intuitive and affordable, we empower local industries to scale rapidly and maintain strategic momentum without relying on flawed or inaccessible foreign software.

# Domain analysis

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## 2.1

### Customer Analysis:

The success of Konvert depends on a granular understanding of the customers who will utilize this engine as their automated backbone. Our analysis identifies two primary enterprise segments and three distinct user archetypes.

#### 2.1.1

### Enterprise Segments:

1. **Large-Scale Pharmaceutical and Distribution Enterprises**

These organizations manage massive operations characterized by an eight-tier hierarchy. Their primary pain point is informational asymmetry—where field data takes days to reach the executive level. For these customers, Konvert provides a unified environment that eliminates data silos and ensures that strategic decisions are based on real-time field intelligence rather than obsolete reports.

2. **Small and Medium Enterprises (SMEs)**

SMEs represent the majority of the Pakistani business landscape but are often excluded from high-tier sales technology due to prohibitive costs. Konvert offers these customers a scalable, localized, and affordable alternative to global giants. This segment values the system's ability to automate mundane tasks, allowing them to grow their operations without a proportional increase in administrative overhead.

### 2.1.2

## User Archetypes:

### 1. Executive Leadership

CEO and C-Suite

Executives require high-level, actionable insights. They use the system to monitor national sales trends, evaluate regional performance, and utilize AI forecasting to plan future market expansions. Their goal is the protection of market share and the optimization of long-term ROI.

### 2. Sales Managers

Tier 2-7

Mid-level managers are responsible for team productivity. They rely on Konvert to dismantle management overload by utilizing automated reporting and real-time oversight. The system allows them to identify underperforming routes and provide data-driven coaching to their teams.

### 3. Field Sales Representatives

Tier 8

The primary generators of data, field representatives require a tool that reduces administrative friction. They utilize the mobile interface for live route tracking, automated tour planning, and inventory management. For this archetype, the value lies in spending less time on paperwork and more time on active sales.

## 2.2

## Stakeholders

The Konvert ecosystem involves several internal and external stakeholders whose collaboration is essential for the technical deployment and operational success of the system.



### 2.2.1

#### Internal Stakeholders:

|                                                                                                                                                                                                                        |                                                                                                                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Project Development Team</b><br>Asim, Obaid, Farheen<br><br>Technical architects responsible for the full-stack engineering, AI model training, and the implementation of the bi-directional synchronization logic. | <b>Project Advisor</b><br>Hafiz Ahsan Arshad<br><br>Strategic consultant and technical validator who oversees the development lifecycle to ensure it adheres to professional software engineering standards and academic requirements. |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 2.2.1

#### External Stakeholders:

|                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Bristol Mayer Biotech</b><br>Abdul Razzaq (Business unit head)<br><br>The lead industry pilot and primary end-user group. Their role is to provide the operational context and act as the primary testing ground for the eight-tier hierarchical visibility. | <b>Almashreqs Soft.</b><br>Bashir Ahmed (CEO)<br><br>Infrastructure partner managing the central web database. Their role is critical in providing secure API endpoints and ensuring the technical compatibility of the web-to-mobile data bridge. |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### Local SME Sector (Optional for FYP):

Secondary stakeholder group acting as market validators. Their role is to provide feedback on the system's affordability and ease of adoption for non-enterprise businesses.

**Note:** The above list is tentative and is subject to change in the future.

## 2.3

# Affected Groups and Social & Economic Impact

The implementation of Konvert generates a ripple effect across various social and economic strata, influencing how labor is utilized and how national capital is retained.

### Social Impact

## Labor Upskilling and Digital Inclusion

Konvert acts as a catalyst for social mobility within the commercial workforce. By automating mundane administrative tasks, it empowers lower-tier field workers to transition from clerical data entry to high-value strategic selling. Furthermore, its intuitive interface promotes digital inclusion, allowing individuals from varied educational backgrounds to navigate complex enterprise systems with minimal training. This reduces the “technical intimidation” barrier, fostering a more agile and digitally-literate workforce in Pakistan.

### Economic Impact

## Capital Retention and National Growth

On a macroeconomic scale, Konvert contributes to Pakistan's economic resilience by promoting technological sovereignty. By providing a domestic alternative to foreign-managed CRM/ERP systems, it prevents the outflow of capital currently spent on international subscription fees. Locally, it empowers SMEs—the engines of national growth—to compete with large-scale distributors by providing them with elite-tier analytical tools at a fraction of the cost, thereby catalyzing broad-market competitiveness and innovation.

### Organizational Impact

## Structural Efficiency

Within an organization, Konvert shifts the culture from reactive error-correction to proactive growth. By eliminating the informational asymmetry that typically plagues multi-tier hierarchies, it reduces internal friction and allows departments to operate in synchronization. This structural efficiency leads to lower operational overhead and higher net profitability, creating a more sustainable business model for Pakistani enterprises.





## 2.4

# Dependencies and External Systems

The functionality of Konvert is contingent upon a robust infrastructure of external services and technical dependencies that provide the specialized capabilities required for enterprise-level sales management.

### 1. Centralized Web Infrastructure (Almashreqs)

The existing web-based business infrastructure managed by Almashreqs serves as the legacy data source. Konvert relies on custom API bridges to pull user hierarchies and push field activity logs to the central Bristol servers.

### 2. Geospatial Services (Google Maps API)

The system integrates the Google Maps Platform for geocoding, real-time salesman tracking, and route distance calculations. This is essential for validating field attendance and optimizing tour plans.

### 3. Machine Learning Frameworks

The AI engine depends on Scikit-learn and TensorFlow for training predictive performance models. These libraries allow the system to process historical field data and generate future sales outlooks.

### 4. Google Firebase Ecosystem

The system utilizes Google Firebase for real-time infrastructure, employing Firestore for NoSQL data synchronization, Cloud Messaging (FCM) for instant notifications, and Firebase Authentication for secure identity management and session control.

### 5. Identity and Security Services

Google Authenticator is integrated to provide Multi-Factor Authentication (MFA), securing executive-level access and sensitive commercial data.

## 2.4

### Related Projects

To accurately position Konvert within the current technological landscape, we conducted a critical analysis of existing solutions currently serving the regional and global markets.

Regional Industry Leader

#### **SanSFE (Regional Industry Leader)**

Primarily utilized within the pharmaceutical distribution sectors of South Asia, SanSFE represents the legacy baseline. While it provides basic field reporting, it suffers from a critical lack of real-time bi-directional synchronization. Data generated on the mobile frontend often requires manual reconciliation with central web servers, leading to significant latency and high error margins in hierarchical reporting.

Global Enterprise CRM Solutions

#### **Salesforce, HubSpot**

These platforms represent the gold standard in sales management. While technically superior, they are largely unsuitable for the Pakistani SME and distribution landscape. Their prohibitive dollar-based subscription costs create a massive financial barrier, and their generalized architectures often require expensive third-party consultants to customize for localized pharmaceutical or multi-tier distribution workflows.

Smaller competition

#### **Localized Legacy ERPs**

Many local enterprises utilize custom-built or older ERP systems. These systems are typically desktop-bound or web-only, lacking the robust mobile integration required for field force tracking. They operate as static record-keeping tools rather than proactive engines, offering no geospatial tracking or AI-driven predictive analytics to assist in route planning or performance forecasting.

## 2.5

# Feature Comparison

| Feature Attribute | SanSFE (Regional)        | Global CRM (Salesforce)   | Requirement Gap                |
|-------------------|--------------------------|---------------------------|--------------------------------|
| Data Sync         | Manual reconciliation.   | High-speed cloud sync.    | Localized bi-directional flow. |
| Mobile UI         | Outdated legacy design.  | Modern but complex.       | Simplified field-force UX.     |
| Reporting         | Static PDF logs.         | Advanced BI dashboards.   | Automated localized metrics.   |
| Cost              | Moderate localized.      | Prohibitive dollar-based. | Affordability for local SME.   |
| Visibility        | Immediate superior only. | Customizable/Complex.     | Native 8-tier hierarchy.       |
| Intelligence      | Record-based only.       | High-cost AI add-ons.     | Built-in native AI engine.     |
| Offline Work      | Poor/Unstable.           | Basic/Cloud-dependent.    | Robust offline-first engine.   |
| Onboarding        | Lengthy setup.           | Long consultant cycles.   | Agile rapid deployment.        |
| Support           | Limited regional.        | Remote/Indirect.          | Direct local engineering.      |

## 2.6

# Reference Documents

## Technical Standards and Quality Framework

- **ISO/IEC 25010:2011:** *Systems and software engineering quality models*. Establishes the framework for evaluating "Product Quality" (8 characteristics like security and maintainability) and "Quality in Use."
- **ISO/IEC/IEEE 12207:2017:** *Software life cycle processes*. A primary standard for governing the full lifespan of a system—from initial concept and development to maintenance and retirement.
- **IEEE Std 830-1998:** *Recommended Practice for Software Requirements Specifications (SRS)*. Used to define system boundaries and ensure requirements are clear, testable, and unambiguous.
- **NIST Special Publication 800-63B:** *Digital Identity Guidelines*. Provides technical requirements for secure authentication, password lifecycle management, and identity proofing.
- **ISO 9241-210:2019:** *Human-centered design for interactive systems*. Focuses on usability and user experience (UX), specifically guiding the interface design for the field-force mobile application.
- **IEEE 1012-2012:** *Standard for System, Software, and Hardware Verification and Validation (V&V)*. Ensures the system is built correctly and satisfies its intended user needs.

## Technical Specifications and Implementation

- **W3C Geolocation API Specification:** *The industry standard for retrieving geographic location information (Latitude/Longitude) via hosting devices.*
- **SWEBOK Guide V3.0:** *Guide to the Software Engineering Body of Knowledge*. Provides a globally recognized boundary for software engineering as a discipline, utilized here for project management and methodology.
- **Fielding, R. T. (2000):** *Architectural Styles and the Design of Network-based Software Architectures*. The foundational doctoral thesis that defined REST (Representational State Transfer), the backbone of modern API design.
- **Google Maps Platform Technical Reference:** *Documentation for implementing high-precision location services, specifically the Geocoding and Distance Matrix APIs for route optimization.*
- **Firebase Realtime Architecture Patterns:** Benchmarks for "offline-first" sync and managing high-concurrency data in NoSQL JSON-tree structures.



## Research and Data Sovereignty

- **Hochreiter, S., & Schmidhuber, J. (1997):** *Long Short-Term Memory (LSTM)*. The seminal paper on Recurrent Neural Networks (RNNs) used for time-series forecasting and sales trend prediction.
- **General Data Privacy Principles (GDPR/APPI):** The legal and conceptual frameworks (EU's GDPR and Japan's APPI) ensuring data sovereignty, residency, and privacy for commercial field-force data.

# Requirement Analysis

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## 3.1 Functional Requirements

The functional requirements define the operational capabilities required for Konvert to serve as a comprehensive sales backbone.

### FR Module 1 Initialization, Security, and Access Control

|       |                                                                                                                                                                                   |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-1. | <b>System Startup Checks</b><br><br>The app must perform security, stability, and update checks during the splash screen sequence before granting access.                         |
| FR-2. | <b>Error Handling and Recovery</b><br><br>The system must provide self-healing options (cache clear, database resync, reinstall) in response to runtime errors.                   |
| FR-3. | <b>Domain and Hierarchy Onboarding</b><br><br>Users must be able to connect to a specific company domain (e.g., Bristol) via API keys and retrieve the specific 8-tier hierarchy. |

|              |                                                                                                                                                            |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>FR-4.</b> | <b>Multi-Factor Authentication (MFA)</b><br><br>Login must be secured via Google Authenticator tokens, validated against the Almashreqs database hash.     |
| <b>FR-5.</b> | <b>Feature Locking Mechanism</b><br><br>The app must restrict feature access dynamically based on the user's specific role within the retrieved hierarchy. |

## FR Module 2

### Data Synchronization and Persistence

|              |                                                                                                                                                                                   |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>FR-6.</b> | <b>Master Sync Protocol</b><br><br>The system must perform a full synchronization with the Almashreq cloud CRM upon login, downloading only data relevant to the specific UserID. |
| <b>FR-7.</b> | <b>Bi-Directional Real-Time Sync</b><br><br>Field data (orders, logs) must sync instantly with Firebase Firestore; web-side updates must push to the mobile app immediately.      |
| <b>FR-8.</b> | <b>Offline-First Architecture</b><br><br>The app must remain fully functional without internet, queueing all transactions for automatic upload upon reconnection.                 |

### FR Module 3

## Intelligent Dashboard and Analytics

|               |                                                                                                                                                                                |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>FR-9.</b>  | <b>AI Performance Prediction</b><br><br>The dashboard must display an AI-generated forecast of the user's weekly success probability based on historical behavioral patterns.  |
| <b>FR-10.</b> | <b>Target Visualization</b><br><br>Real-time graphs must show daily and weekly target achievement progress derived from the active tour plan.                                  |
| <b>FR-11.</b> | <b>Automated Notification System</b><br><br>A background checker must fetch and display prioritized alerts (warnings, updates, approvals) via a dedicated notification center. |
| <b>FR-12.</b> | <b>Advanced Reporting Module</b><br><br>The system must generate detailed performance reports with granular filtering (daily, weekly, categorical) using cloud reporting APIs. |

### FR Module 4

## Tour Planning and Roadmap

|               |                                                                                                                                         |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>FR-13.</b> | <b>Automated Tour Plan Creation</b><br><br>Users must be able to create new tour plans by selecting dates, routes, and expense budgets. |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------|



|               |                                                                                                                                                                            |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>FR-14.</b> | <b>AI Enhancement ("Enhance Button")</b><br><br>The system must offer an AI-helper class to optimize the tour plan for maximum efficiency before submission.               |
| <b>FR-15.</b> | <b>Management Approval Workflow</b><br><br>Submitted tour plans must lock upon creation and unlock only after receiving digital approval signals from the management tier. |

## FR Module 5

### Fieldwork and Geospatial Operations

|               |                                                                                                                                                                         |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>FR-16.</b> | <b>Integrated Route Navigation</b><br><br>The fieldwork interface must embed Google Maps to show the route to the next customer without external app switching.         |
| <b>FR-17.</b> | <b>Geo-Fencing Triggers</b><br><br>Product selection buttons must only become active when the user's GPS coordinates enter a defined radius of the customer's location. |
| <b>FR-18.</b> | <b>Quick-Update Mechanism</b><br><br>The fieldwork page must allow for rapid, localized data fetching from the server without requiring a full Master Sync.             |

## Order Management and Booking

|               |                                                                                                                                                                |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>FR-19.</b> | <b>Dynamic Product Selection</b><br><br>The system must allow users to select OTC and Regular products, applying discounts and bonuses based on current logic. |
| <b>FR-20.</b> | <b>Daily Booking Summary</b><br><br>Users must be able to review, edit, and confirm daily bookings before final submission.                                    |
| <b>FR-21.</b> | <b>End-of-Day Upload</b><br><br>The system must compile all daily bookings and upload them to the Almarshreqs cloud CRM for final administrative approval.     |

### 3.2

## Non-Functional Requirements

The non-functional requirements define the quality attributes, design constraints, and performance benchmarks of the system.

### NFR Module 1

#### Performance and Efficiency

|               |                                                                                                                                                                                             |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NFR-1.</b> | <b>Rendering and Sync Speed</b><br><br>The Flutter UI must maintain 60 FPS for smooth navigation, and the initial Master Sync must complete within 15 seconds on a standard 4G connection.  |
| <b>NFR-2.</b> | <b>Battery Optimization</b><br><br>Background services, including continuous 5-minute GPS polling and AI data gathering, must consume less than 15% of device battery over an 8-hour shift. |

### NFR Module 2

#### Reliability and Availability

|               |                                                                                                                                                                                  |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NFR-3.</b> | <b>Offline Resilience</b><br><br>The system must guarantee zero data loss during network dropouts by utilizing robust local caching before attempting Firebase synchronization.  |
| <b>NFR-4.</b> | <b>High Availability</b><br><br>The cloud infrastructure and Almashreqs API bridges must maintain a 99.9% uptime SLA to ensure uninterrupted executive monitoring and data flow. |

## NFR Module 3

### Security and Compliance

|               |                                                                                                                                                                                |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NFR-5.</b> | <b>Encryption Standards</b><br><br>All data in transit must be secured via TLS 1.3, and local cache data (like customer PII and bookings) must be encrypted using AES-256.     |
| <b>NFR-6.</b> | <b>Breach Management</b><br><br>The system must auto-expire idle sessions and instantly wipe local data while alerting admins if a device security breach or root is detected. |

## NFR Module 4

### Scalability and Maintainability

|               |                                                                                                                                                                                                |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NFR-7.</b> | <b>Concurrent Architecture</b><br><br>The Firebase NoSQL structure and API bridges must be optimized to handle 5,000+ simultaneous active field agents without write-lock bottlenecks.         |
| <b>NFR-8.</b> | <b>Application Updates</b><br><br>The system must support forced mandatory updates and optional delayed updates (1-2 days) to ensure all tiers operate on the same compliant software version. |

## NFR Module 5

### Usability and Compatibility

|                |                                                                                                                                                                                     |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NFR-9.</b>  | <b>Hardware Compatibility</b><br><br>The application must operate efficiently on lower-tier mobile devices (minimum 2GB RAM) typically deployed to regional field representatives.  |
| <b>NFR-10.</b> | <b>Human-Centered UX</b><br><br>Adhering to ISO 9241-210 standards, core field workflows (check-in, booking) must be achievable within three screen taps to minimize user friction. |

### 3.3

## List of Actors

The Konvert system interacts with the following stakeholders across the enterprise landscape:

- Executive Leadership (CEO / Business Unit Head)
- Tactical Managers (National / Regional / Zonal / Area)
- Operations Support (Command Center Operatives)
- Field Sales Representatives (Tier 8)
- System Administrator

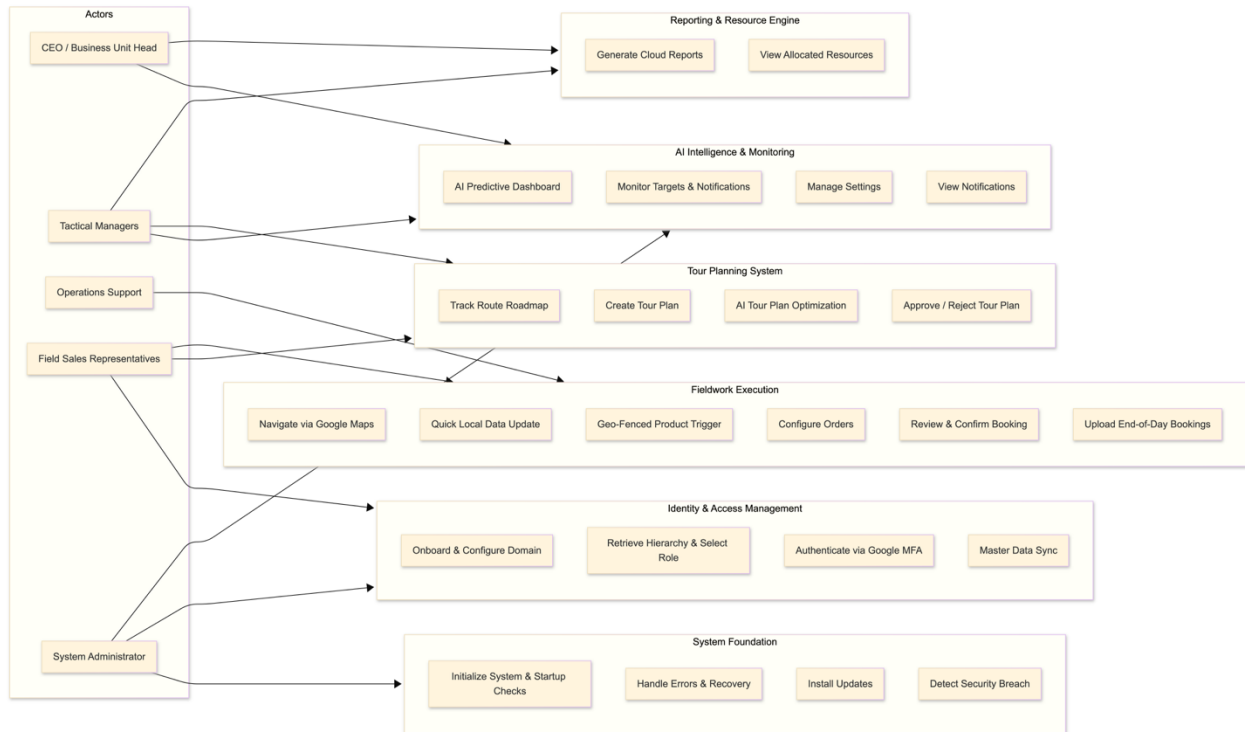
### 3.4

## List of Use Cases

|       |                                                          |
|-------|----------------------------------------------------------|
| UC-01 | Initialize System and Perform Application Startup Checks |
| UC-02 | Handle Application Errors and Execute Recovery Options   |
| UC-03 | Download and Install Mandatory or Optional Updates       |
| UC-04 | Detect and Report System Security Breaches               |
| UC-05 | Onboard User and Configure Enterprise Domain Connection  |
| UC-06 | Retrieve Hierarchy and Select Business Role              |
| UC-07 | Authenticate User via Google Authenticator MFA           |
| UC-08 | Execute User-Specific Master Data Synchronization        |
| UC-09 | View AI-Powered Predictive Performance Dashboard         |
| UC-10 | Monitor Target Goals, Promotions, and Notifications      |
| UC-11 | Manage User Profile, Theme, and Application Settings     |
| UC-12 | View and Filter Prioritized System Notifications         |
| UC-13 | Generate and Filter Advanced Cloud-Based Reports         |
| UC-14 | View and Filter Allocated Enterprise Resources           |
| UC-15 | Track and Manage Current Route Roadmap                   |
| UC-16 | Create and Configure New Route Tour Plan                 |
| UC-17 | Optimize Tour Plan via AI Enhancement Tool               |
| UC-18 | Approve or Reject Submitted Tour Plans                   |
| UC-19 | Navigate Assigned Route via Integrated Google Maps       |
| UC-20 | Execute Quick Data Update for Localized Assets           |
| UC-21 | Trigger Geo-Fenced Customer Product Selection            |
| UC-22 | Select Products, Discounts, and Configure Orders         |
| UC-23 | Review, Edit, and Confirm Customer Bookings              |
| UC-24 | Upload End-of-Day Bookings to Central Cloud CRM          |

### 3.5

## Use Case Diagram



For detailed representation [Click Here](#)

### 3.6

## Extended Use Case

UC-01

### Initialize System and Perform Application Startup Checks

| Field                    | Description                                                                                                                                                                                                                                                                                                                               |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID              | UC-01                                                                                                                                                                                                                                                                                                                                     |
| Name                     | Initialize System and Perform Application Startup Checks                                                                                                                                                                                                                                                                                  |
| Primary Actor            | Sales Representative (Business User)                                                                                                                                                                                                                                                                                                      |
| Supporting Actors        | Almashreq CRM API, Firebase, Update Server                                                                                                                                                                                                                                                                                                |
| Preconditions            | App is installed on device                                                                                                                                                                                                                                                                                                                |
| Trigger                  | User launches the Konvert mobile app                                                                                                                                                                                                                                                                                                      |
| Main Flow                | 1. Display splash screen animation. 2. Perform app integrity check. 3. Perform update availability check. 4. Perform security status validation. 5. Validate stored login token (if exists). 6. Check pending sync requirements. 7. Redirect user to appropriate next page (Login / Master Sync / Dashboard / Update / Error / Security). |
| Alternate Flows          | A1: If update mandatory → Redirect to Update Page. A2: If security breach detected → Redirect to Security Page. A3: If data mismatch → Redirect to Master Sync. A4: If first launch → Redirect to Onboarding. A5: If startup error occurs → Redirect to Error Page.                                                                       |
| Postconditions (Success) | App transitions to appropriate operational state.                                                                                                                                                                                                                                                                                         |
| Postconditions (Failure) | User is shown error recovery options.                                                                                                                                                                                                                                                                                                     |



|                       |                                                                     |
|-----------------------|---------------------------------------------------------------------|
| <b>Business Rules</b> | Startup validation must complete within performance limits (NFR-1). |
|-----------------------|---------------------------------------------------------------------|

UC-02

## Handle Application Errors and Execute Recovery Options

| Field                           | Description                                                                                                                                                                        |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>              | UC-02                                                                                                                                                                              |
| <b>Name</b>                     | Handle Application Errors and Execute Recovery Options                                                                                                                             |
| <b>Primary Actor</b>            | Sales Representative                                                                                                                                                               |
| <b>Supporting Actors</b>        | Local Storage Engine, Update Server                                                                                                                                                |
| <b>Preconditions</b>            | System detects startup or runtime failure                                                                                                                                          |
| <b>Trigger</b>                  | Error condition raised by system                                                                                                                                                   |
| <b>Main Flow</b>                | 1. Display error details to user. 2. Provide contextual recovery options: • Restart App • Clear Cache • Clear Local Database • Resync Data • Update App • Reinstall via Play Store |
| <b>Alternate Flows</b>          | A1: If cache clear resolves issue → Restart to Splash. A2: If database cleared → Redirect to Master Sync. A3: If update required → Redirect to Update Page.                        |
| <b>Postconditions (Success)</b> | App restored to stable working condition.                                                                                                                                          |
| <b>Postconditions (Failure)</b> | User instructed to reinstall application.                                                                                                                                          |
| <b>Business Rules</b>           | Zero corruption tolerance for synced CRM data (NFR-3).                                                                                                                             |

UC-03

## Download and Install Mandatory or Optional Updates

| Field                       | Description                                                                                                                                                                               |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID                 | UC-03                                                                                                                                                                                     |
| Name                        | Download and Install Mandatory or Optional Updates                                                                                                                                        |
| Primary Actor               | Sales Representative                                                                                                                                                                      |
| Supporting Actors           | Update Server, Google Play API                                                                                                                                                            |
| Preconditions               | Update detected during startup or background check                                                                                                                                        |
| Trigger                     | System flags available update                                                                                                                                                             |
| Main Flow                   | 1. Display update information (version, importance). 2. If mandatory → Begin automatic download. 3. Lock app access until update completes. 4. Restart app after successful installation. |
| Alternate Flows             | A1: If optional → Allow user 1–2 day delay. A2: If download fails → Show retry option.                                                                                                    |
| Postconditions<br>(Success) | Application updated and relaunched.                                                                                                                                                       |
| Postconditions<br>(Failure) | App remains unusable until mandatory update completes.                                                                                                                                    |
| Business Rules              | Version consistency required across hierarchy tiers (NFR-8).                                                                                                                              |

UC-04

## Detect and Report System Security Breaches

| Field       | Description                                |
|-------------|--------------------------------------------|
| Use Case ID | UC-04                                      |
| Name        | Detect and Report System Security Breaches |



|                                 |                                                                                                                                                                                                                      |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Primary Actor</b>            | System (Automated Security Engine)                                                                                                                                                                                   |
| <b>Supporting Actors</b>        | System Admin, Almashreq CRM Security API                                                                                                                                                                             |
| <b>Preconditions</b>            | App running in foreground or background                                                                                                                                                                              |
| <b>Trigger</b>                  | Security anomaly detected (root access, tampering, unauthorized modification)                                                                                                                                        |
| <b>Main Flow</b>                | 1. Detect abnormal security condition. 2. Validate breach authenticity. 3. Auto-expire active sessions. 4. Wipe encrypted local data (if severe). 5. Send breach report to admin. 6. Redirect user to Security Page. |
| <b>Alternate Flows</b>          | A1: Minor warning → Notify user but allow limited usage. A2: Severe breach → Lock app completely.                                                                                                                    |
| <b>Postconditions (Success)</b> | System integrity preserved.                                                                                                                                                                                          |
| <b>Postconditions (Failure)</b> | App access denied.                                                                                                                                                                                                   |
| <b>Business Rules</b>           | Must comply with TLS 1.3 & AES-256 encryption standards (NFR-5).                                                                                                                                                     |

UC-05

## Onboard User and Configure Enterprise Domain Connection

| Field                    | Description                                             |
|--------------------------|---------------------------------------------------------|
| <b>Use Case ID</b>       | UC-05                                                   |
| <b>Name</b>              | Onboard User and Configure Enterprise Domain Connection |
| <b>Primary Actor</b>     | Sales Representative                                    |
| <b>Supporting Actors</b> | Almashreq CRM API                                       |
| <b>Preconditions</b>     | App launched for first time                             |

|                                 |                                                                                                                                                                                                                                                                                                            |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Trigger</b>                  | First-time app access                                                                                                                                                                                                                                                                                      |
| <b>Main Flow</b>                | 1. Collect user preferences (theme, language). 2. Display app introduction modal. 3. Present Terms of Service agreement. 4. User enters domain address + API key. 5. Validate domain with Almashreq CRM. 6. Retrieve company hierarchy structure. 7. Allow user to select role. 8. Redirect to Login Page. |
| <b>Alternate Flows</b>          | A1: Invalid API key → Display error and retry option. A2: Domain unreachable → Show network error and retry.                                                                                                                                                                                               |
| <b>Postconditions (Success)</b> | App configured for specific enterprise instance.                                                                                                                                                                                                                                                           |
| <b>Postconditions (Failure)</b> | User cannot proceed to login.                                                                                                                                                                                                                                                                              |
| <b>Business Rules</b>           | Hierarchy determines feature-lock configuration (FR-5).                                                                                                                                                                                                                                                    |

UC-06

## Retrieve Hierarchy and Select Business Role

| Field                    | Description                                        |
|--------------------------|----------------------------------------------------|
| <b>Use Case ID</b>       | UC-06                                              |
| <b>Name</b>              | Retrieve Hierarchy and Select Business Role        |
| <b>Primary Actor</b>     | Sales Representative                               |
| <b>Supporting Actors</b> | Almashreq CRM API                                  |
| <b>Preconditions</b>     | Valid domain + API key submitted (UC-05 completed) |
| <b>Trigger</b>           | Successful enterprise domain validation            |

|                                 |                                                                                                                                                                                                                                                                                                                          |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Main Flow</b>                | 1. System requests hierarchy structure from Almashreq CRM. 2. CRM returns 8-tier business hierarchy. 3. System checks if customer module enabled (v1.0 disabled). 4. Display role selection interface. 5. User selects their business role. 6. System stores selected hierarchy role locally. 7. Redirect to Login Page. |
| <b>Alternate Flows</b>          | A1: Hierarchy fetch fails → Show retry option. A2: No role available → Deny access.                                                                                                                                                                                                                                      |
| <b>Postconditions (Success)</b> | Role stored and feature-lock rules prepared.                                                                                                                                                                                                                                                                             |
| <b>Postconditions (Failure)</b> | Onboarding halted.                                                                                                                                                                                                                                                                                                       |
| <b>Business Rules</b>           | Role determines feature access via Feature Lock Engine (FR-5).                                                                                                                                                                                                                                                           |

## UC-07

### Authenticate User via Google Authenticator MFA

| Field                    | Description                                                                                                                               |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>       | UC-07                                                                                                                                     |
| <b>Name</b>              | Authenticate User via Google Authenticator MFA                                                                                            |
| <b>Primary Actor</b>     | Sales Representative                                                                                                                      |
| <b>Supporting Actors</b> | Almashreq Database, Google Authenticator API                                                                                              |
| <b>Preconditions</b>     | Role selected, login page active                                                                                                          |
| <b>Trigger</b>           | User submits login credentials                                                                                                            |
| <b>Main Flow</b>         | 1. Validate username/password with Almashreq database. 2. If valid → Prompt MFA token. 3. User enters Google Authenticator OTP. 4. System |

|                                 |                                                                                                                                     |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
|                                 | verifies OTP hash against database. 5. Generate secure session token. 6. Store encrypted token locally. 7. Redirect to Master Sync. |
| <b>Alternate Flows</b>          | A1: Invalid credentials → Show error. A2: Invalid OTP → Retry limit enforced. A3: Session token mismatch → Force re-login.          |
| <b>Postconditions (Success)</b> | Authenticated session established.                                                                                                  |
| <b>Postconditions (Failure)</b> | Access denied.                                                                                                                      |
| <b>Business Rules</b>           | MFA mandatory for all business-tier users (FR-4, NFR-5).                                                                            |

UC-08

## Execute User-Specific Master Data Synchronization

| Field                    | Description                                                                                                                                                                                                                          |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>       | UC-08                                                                                                                                                                                                                                |
| <b>Name</b>              | Execute User-Specific Master Data Synchronization                                                                                                                                                                                    |
| <b>Primary Actor</b>     | Sales Representative                                                                                                                                                                                                                 |
| <b>Supporting Actors</b> | Almashreq CRM API, Firebase Firestore                                                                                                                                                                                                |
| <b>Preconditions</b>     | Authenticated session exists                                                                                                                                                                                                         |
| <b>Trigger</b>           | Successful login or manual sync                                                                                                                                                                                                      |
| <b>Main Flow</b>         | 1. Initiate secure cloud connection. 2. Fetch data limited to current UserID. 3. Download customers, resources, targets, tour plan, permissions. 4. Encrypt and cache locally. 5. Validate data integrity. 6. Redirect to Dashboard. |
| <b>Alternate Flows</b>   | A1: Network failure → Retry or enable offline mode. A2: Data corruption → Redirect to Error Page.                                                                                                                                    |

|                                 |                                                              |
|---------------------------------|--------------------------------------------------------------|
| <b>Postconditions (Success)</b> | Local database synced and ready for offline-first operation. |
| <b>Postconditions (Failure)</b> | User restricted until sync completes.                        |
| <b>Business Rules</b>           | Master Sync must complete within 15 seconds (NFR-1).         |

UC-09

## View AI-Powered Predictive Performance Dashboard

| Field                           | Description                                                                                                                                            |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>              | UC-09                                                                                                                                                  |
| <b>Name</b>                     | View AI-Powered Predictive Performance Dashboard                                                                                                       |
| <b>Primary Actor</b>            | Sales Representative                                                                                                                                   |
| <b>Supporting Actors</b>        | AI Analytics Engine, Reporting API                                                                                                                     |
| <b>Preconditions</b>            | Master Sync completed                                                                                                                                  |
| <b>Trigger</b>                  | User opens Dashboard Home Tab                                                                                                                          |
| <b>Main Flow</b>                | 1. Load weekly progress data. 2. Run AI behavioral analysis model. 3. Generate success probability forecast. 4. Display prediction graph and insights. |
| <b>Alternate Flows</b>          | A1: AI model unavailable → Display historical performance only.                                                                                        |
| <b>Postconditions (Success)</b> | User views predictive performance insights.                                                                                                            |
| <b>Postconditions (Failure)</b> | AI disabled, dashboard still functional.                                                                                                               |
| <b>Business Rules</b>           | AI forecast must be derived from company dataset (FR-9).                                                                                               |

## UC-10

### Monitor Target Goals, Promotions, and Notifications

| Field             | Description                                                                                                                                                 |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID       | UC-10                                                                                                                                                       |
| Name              | Monitor Target Goals, Promotions, and Notifications                                                                                                         |
| Primary Actor     | Sales Representative                                                                                                                                        |
| Supporting Actors | Notification Engine, Promotion API                                                                                                                          |
| Preconditions     | Dashboard active                                                                                                                                            |
| Trigger           | User opens Home Tab                                                                                                                                         |
| Main Flow         | 1. Display daily and weekly targets. 2. Fetch active promotion banner. 3. Display latest prioritized notification. 4. Show optional ad banner (if enabled). |
| Alternate Flows   | A1: Promotion unavailable → Hide banner. A2: No notifications → Show placeholder.                                                                           |
| Postconditions    | User informed of operational priorities.                                                                                                                    |
| Business Rules    | Targets must match approved Tour Plan data.                                                                                                                 |

## UC-11

### Manage User Profile, Theme, and Application Settings

| Field         | Description                                          |
|---------------|------------------------------------------------------|
| Use Case ID   | UC-11                                                |
| Name          | Manage User Profile, Theme, and Application Settings |
| Primary Actor | Sales Representative                                 |



|                          |                                                                                                                                                        |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Supporting Actors</b> | Local Settings Manager                                                                                                                                 |
| <b>Preconditions</b>     | User logged in                                                                                                                                         |
| <b>Trigger</b>           | User opens Settings Page                                                                                                                               |
| <b>Main Flow</b>         | 1. Display profile details. 2. Allow theme and language changes. 3. Toggle automation features. 4. Save preferences locally. 5. Provide logout option. |
| <b>Alternate Flows</b>   | A1: Logout selected → Clear token and redirect to Login.                                                                                               |
| <b>Postconditions</b>    | Updated preferences stored and applied instantly.                                                                                                      |
| <b>Business Rules</b>    | Session token must be wiped upon logout (NFR-6).                                                                                                       |

UC-12

## View and Filter Prioritized System Notifications

| Field                    | Description                                                                                                                              |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>       | UC-12                                                                                                                                    |
| <b>Name</b>              | View and Filter Prioritized System Notifications                                                                                         |
| <b>Primary Actor</b>     | Sales Representative                                                                                                                     |
| <b>Supporting Actors</b> | Notification Engine, Firebase                                                                                                            |
| <b>Preconditions</b>     | User logged in                                                                                                                           |
| <b>Trigger</b>           | User opens Notification Page                                                                                                             |
| <b>Main Flow</b>         | 1. Fetch categorized notifications. 2. Sort by priority (High / Medium / Low). 3. Allow filtering options. 4. Mark notification as read. |
| <b>Alternate Flows</b>   | A1: Offline mode → Show cached notifications only.                                                                                       |
| <b>Postconditions</b>    | Notifications updated and synced.                                                                                                        |

|                       |                                                    |
|-----------------------|----------------------------------------------------|
| <b>Business Rules</b> | Real-time updates must be supported (FR-11, FR-7). |
|-----------------------|----------------------------------------------------|

## UC-13

# Generate and Filter Advanced Cloud-Based Reports

| Field                           | Description                                                                                                                                                                                                                                                                                         |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>              | UC-13                                                                                                                                                                                                                                                                                               |
| <b>Name</b>                     | Generate and Filter Advanced Cloud-Based Reports                                                                                                                                                                                                                                                    |
| <b>Primary Actor</b>            | Sales Representative                                                                                                                                                                                                                                                                                |
| <b>Supporting Actors</b>        | Almashreq Reporting API, Chart/Visualization API                                                                                                                                                                                                                                                    |
| <b>Preconditions</b>            | User authenticated and Master Sync completed                                                                                                                                                                                                                                                        |
| <b>Trigger</b>                  | User opens Reporting Tab                                                                                                                                                                                                                                                                            |
| <b>Main Flow</b>                | 1. System establishes secure API connection. 2. Fetch user-specific reporting dataset. 3. Display summary KPIs. 4. Provide filtering options (daily, weekly, monthly, category, product, region). 5. Render charts dynamically using reporting APIs. 6. Update data view based on selected filters. |
| <b>Alternate Flows</b>          | A1: Network unavailable → Show last cached report snapshot. A2: API error → Display error and retry option.                                                                                                                                                                                         |
| <b>Postconditions (Success)</b> | User views detailed performance analytics.                                                                                                                                                                                                                                                          |
| <b>Postconditions (Failure)</b> | Reporting limited to cached data.                                                                                                                                                                                                                                                                   |
| <b>Business Rules</b>           | Reports must reflect real-time cloud data consistency (FR-12, FR-7).                                                                                                                                                                                                                                |

## UC-14

### View and Filter Allocated Enterprise Resources

| Field             | Description                                                                                                                                                                                                                          |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID       | UC-14                                                                                                                                                                                                                                |
| Name              | View and Filter Allocated Enterprise Resources                                                                                                                                                                                       |
| Primary Actor     | Sales Representative                                                                                                                                                                                                                 |
| Supporting Actors | Almashreq CRM API                                                                                                                                                                                                                    |
| Preconditions     | Master Sync completed                                                                                                                                                                                                                |
| Trigger           | User opens Resource Tab                                                                                                                                                                                                              |
| Main Flow         | 1. Load locally cached resource dataset. 2. Display categorized resources (customers, products, bricks, locations). 3. Allow advanced filtering (area, product type, status). 4. Display detailed resource information on selection. |
| Alternate Flows   | A1: Data outdated → Prompt quick update (UC-20).                                                                                                                                                                                     |
| Postconditions    | User accesses allocated operational assets.                                                                                                                                                                                          |
| Business Rules    | User can only view resources assigned to their UserID (FR-6).                                                                                                                                                                        |

## UC-15

### Track and Manage Current Route Roadmap

| Field             | Description                            |
|-------------------|----------------------------------------|
| Use Case ID       | UC-15                                  |
| Name              | Track and Manage Current Route Roadmap |
| Primary Actor     | Sales Representative                   |
| Supporting Actors | Local Database, Tour Plan Module       |



|                        |                                                                                                                                                                                  |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Preconditions</b>   | Approved tour plan exists                                                                                                                                                        |
| <b>Trigger</b>         | User opens Roadmap Tab                                                                                                                                                           |
| <b>Main Flow</b>       | 1. Retrieve current approved tour plan. 2. Display roadmap timeline (Completed / In Progress / Upcoming). 3. Show daily targets tied to route stops. 4. Highlight next customer. |
| <b>Alternate Flows</b> | A1: No tour plan found → Prompt to create new plan (UC-16).                                                                                                                      |
| <b>Postconditions</b>  | User has clear visibility of route execution progress.                                                                                                                           |
| <b>Business Rules</b>  | Only approved tour plans are actionable.                                                                                                                                         |

UC-16

## Create and Configure New Route Tour Plan

| Field                    | Description                                                                                                                                                                                                                                              |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>       | UC-16                                                                                                                                                                                                                                                    |
| <b>Name</b>              | Create and Configure New Route Tour Plan                                                                                                                                                                                                                 |
| <b>Primary Actor</b>     | Sales Representative                                                                                                                                                                                                                                     |
| <b>Supporting Actors</b> | Resource Module, Management Approval System                                                                                                                                                                                                              |
| <b>Preconditions</b>     | No active approved tour plan OR update request approved                                                                                                                                                                                                  |
| <b>Trigger</b>           | User selects "Create / Update Tour Plan"                                                                                                                                                                                                                 |
| <b>Main Flow</b>         | 1. Load allocated resources. 2. User selects dates, timing, routes, customers. 3. Define daily targets and expenses. 4. Validate route constraints. 5. Save draft tour plan. 6. Submit to management for approval. 7. Lock plan until approval response. |
| <b>Alternate Flows</b>   | A1: Validation error → Highlight incorrect inputs. A2: User cancels → Discard draft.                                                                                                                                                                     |

|                       |                                                                                    |
|-----------------------|------------------------------------------------------------------------------------|
| <b>Postconditions</b> | Tour plan submitted and pending approval.                                          |
| <b>Business Rules</b> | Plan must align with allocated resources and hierarchy permissions (FR-13, FR-15). |

UC-17

## Optimize Tour Plan via AI Enhancement Tool

| Field                    | Description                                                                                                                                                                                                                                            |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>       | UC-17                                                                                                                                                                                                                                                  |
| <b>Name</b>              | Optimize Tour Plan via AI Enhancement Tool                                                                                                                                                                                                             |
| <b>Primary Actor</b>     | Sales Representative                                                                                                                                                                                                                                   |
| <b>Supporting Actors</b> | AI TourPlanHelper Engine                                                                                                                                                                                                                               |
| <b>Preconditions</b>     | Tour plan draft exists                                                                                                                                                                                                                                 |
| <b>Trigger</b>           | User presses "Enhance" button                                                                                                                                                                                                                          |
| <b>Main Flow</b>         | 1. Send draft tour plan data to AI engine. 2. Analyze route efficiency, workload balance, target distribution. 3. Suggest optimized customer order and expense adjustments. 4. Present optimization preview. 5. User accepts or rejects modifications. |
| <b>Alternate Flows</b>   | A1: AI unavailable → Disable enhancement feature.                                                                                                                                                                                                      |
| <b>Postconditions</b>    | Optimized tour plan saved (if accepted).                                                                                                                                                                                                               |
| <b>Business Rules</b>    | AI must operate within allocated resource constraints (FR-14).                                                                                                                                                                                         |

UC-18

## Approve or Reject Submitted Tour Plans

| Field                         | Description                                                                                                                                                                                              |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID                   | UC-18                                                                                                                                                                                                    |
| Name                          | Approve or Reject Submitted Tour Plans                                                                                                                                                                   |
| Primary Actor                 | Management Tier (Area/Regional Manager)                                                                                                                                                                  |
| Supporting Actors             | Notification Engine, Tour Plan Module                                                                                                                                                                    |
| Preconditions                 | Tour plan submitted (UC-16 completed)                                                                                                                                                                    |
| Trigger                       | Manager reviews submitted plan                                                                                                                                                                           |
| Main Flow                     | 1. Manager views tour plan details. 2. Evaluate targets, route logic, expenses. 3. Approve or Reject digitally. 4. If approved → Lock plan and notify user. 5. If rejected → Send feedback note to user. |
| Alternate Flows               | A1: Manager requests modification → Return editable state to user.                                                                                                                                       |
| Postconditions<br>(Approval)  | Tour plan becomes active and immutable.                                                                                                                                                                  |
| Postconditions<br>(Rejection) | User notified and allowed to revise.                                                                                                                                                                     |
| Business Rules                | Approval required before roadmap execution begins (FR-15).                                                                                                                                               |

UC-19

## Navigate Assigned Route via Integrated Google Maps

| Field         | Description                                        |
|---------------|----------------------------------------------------|
| Use Case ID   | UC-19                                              |
| Name          | Navigate Assigned Route via Integrated Google Maps |
| Primary Actor | Sales Representative                               |



|                          |                                                                                                                                                                                                          |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Supporting Actors</b> | Google Maps API, GPS Module                                                                                                                                                                              |
| <b>Preconditions</b>     | Approved tour plan exists (UC-18 completed)                                                                                                                                                              |
| <b>Trigger</b>           | User presses "Start Field Work"                                                                                                                                                                          |
| <b>Main Flow</b>         | 1. Retrieve next customer from active roadmap. 2. Fetch GPS coordinates. 3. Embed Google Maps view within Fieldwork page. 4. Display optimized route and distance. 5. Continuously update live position. |
| <b>Alternate Flows</b>   | A1: GPS disabled → Prompt user to enable location services. A2: Network unavailable → Display offline map with limited functionality.                                                                    |
| <b>Postconditions</b>    | User actively navigates to next assigned customer.                                                                                                                                                       |
| <b>Business Rules</b>    | Route must align strictly with approved tour plan.                                                                                                                                                       |

## UC-20

### Execute Quick Data Update for Localized Assets

| Field                    | Description                                                                                                                                                                        |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>       | UC-20                                                                                                                                                                              |
| <b>Name</b>              | Execute Quick Data Update for Localized Assets                                                                                                                                     |
| <b>Primary Actor</b>     | Sales Representative                                                                                                                                                               |
| <b>Supporting Actors</b> | Almashreq CRM API                                                                                                                                                                  |
| <b>Preconditions</b>     | User on Fieldwork Page                                                                                                                                                             |
| <b>Trigger</b>           | User presses "Update" button                                                                                                                                                       |
| <b>Main Flow</b>         | 1. Send request for selective resource refresh. 2. Fetch updated product pricing, targets, or customer data. 3. Merge updated data into local cache. 4. Confirm update completion. |

|                        |                                                    |
|------------------------|----------------------------------------------------|
| <b>Alternate Flows</b> | A1: Network failure → Retry option.                |
| <b>Postconditions</b>  | Localized data refreshed without full Master Sync. |
| <b>Business Rules</b>  | Must not trigger full data reload (FR-18).         |

UC-21

## Trigger Geo-Fenced Customer Product Selection

| Field                    | Description                                                                                                                                                                            |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>       | UC-21                                                                                                                                                                                  |
| <b>Name</b>              | Trigger Geo-Fenced Customer Product Selection                                                                                                                                          |
| <b>Primary Actor</b>     | Sales Representative                                                                                                                                                                   |
| <b>Supporting Actors</b> | GPS Module, Geo-Fencing Engine                                                                                                                                                         |
| <b>Preconditions</b>     | User navigating to customer location                                                                                                                                                   |
| <b>Trigger</b>           | User enters predefined radius around customer                                                                                                                                          |
| <b>Main Flow</b>         | 1. Continuously monitor GPS location. 2. Compare current coordinates with customer radius. 3. If within range → Activate Product Selection button. 4. Allow transition to Product Tab. |
| <b>Alternate Flows</b>   | A1: Outside radius → Button remains disabled. A2: GPS manipulation detected → Trigger security warning.                                                                                |
| <b>Postconditions</b>    | Product selection access granted only within valid location.                                                                                                                           |
| <b>Business Rules</b>    | Geo-fencing mandatory before booking (FR-17).                                                                                                                                          |



## UC-22

### Select Products, Discount, and Configure Orders

| Field             | Description                                                                                                                                                                                                                                                                      |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case ID       | UC-22                                                                                                                                                                                                                                                                            |
| Name              | Select Products, Discounts, and Configure Orders                                                                                                                                                                                                                                 |
| Primary Actor     | Sales Representative                                                                                                                                                                                                                                                             |
| Supporting Actors | Local Product Engine                                                                                                                                                                                                                                                             |
| Preconditions     | Geo-fence validated (UC-21 completed)                                                                                                                                                                                                                                            |
| Trigger           | User enters Product Selection Tab                                                                                                                                                                                                                                                |
| Main Flow         | 1. Display OTC products. 2. User selects quantity, discount, bonus. 3. Add products to temporary cart. 4. Display running total against target. 5. Allow editing/removal of selected items. 6. Validate minimum acceptable target threshold. 7. Enable "Confirm Booking" button. |
| Alternate Flows   | A1: Target not met → Warn user but allow override (if policy permits).                                                                                                                                                                                                           |
| Postconditions    | Order ready for confirmation.                                                                                                                                                                                                                                                    |
| Business Rules    | Discounts and bonuses must comply with cloud-defined logic (FR-19).                                                                                                                                                                                                              |

## UC-23

### Review, Edit, and Confirm Customer Bookings

| Field         | Description                                 |
|---------------|---------------------------------------------|
| Use Case ID   | UC-23                                       |
| Name          | Review, Edit, and Confirm Customer Bookings |
| Primary Actor | Sales Representative                        |

|                          |                                                                                                                                                                            |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Supporting Actors</b> | Local Booking Module                                                                                                                                                       |
| <b>Preconditions</b>     | Products selected                                                                                                                                                          |
| <b>Trigger</b>           | User confirms booking                                                                                                                                                      |
| <b>Main Flow</b>         | 1. Store booking locally in encrypted database. 2. Display booking summary in Bookings Page. 3. Allow editing of individual bookings. 4. Update running total for the day. |
| <b>Alternate Flows</b>   | A1: User edits booking → Return to Product Tab.                                                                                                                            |
| <b>Postconditions</b>    | Booking safely stored for end-of-day upload.                                                                                                                               |
| <b>Business Rules</b>    | Offline-first guarantee applies (NFR-3).                                                                                                                                   |

UC-24

## Upload End-of-Day Bookings to Central Cloud CRM

| Field                    | Description                                                                                                                                                                                                       |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use Case ID</b>       | UC-24                                                                                                                                                                                                             |
| <b>Name</b>              | Upload End-of-Day Bookings to Central Cloud CRM                                                                                                                                                                   |
| <b>Primary Actor</b>     | Sales Representative                                                                                                                                                                                              |
| <b>Supporting Actors</b> | Almashreq CRM API, Firebase Sync Engine                                                                                                                                                                           |
| <b>Preconditions</b>     | At least one booking exists                                                                                                                                                                                       |
| <b>Trigger</b>           | User presses "Confirm End-of-Day Booking"                                                                                                                                                                         |
| <b>Main Flow</b>         | 1. Compile all daily bookings. 2. Encrypt payload via TLS 1.3. 3. Upload to Almashreq cloud CRM. 4. Await acknowledgment response. 5. Mark bookings as submitted. 6. Notify system admin for review and approval. |

|                                     |                                                                                                                            |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>Alternate Flows</b>              | A1: Network failure → Queue upload and retry automatically. A2: Upload rejected → Display rejection note in notifications. |
| <b>Postconditions<br/>(Success)</b> | Bookings successfully transmitted to central system.                                                                       |
| <b>Postconditions<br/>(Failure)</b> | Data retained locally until successful upload.                                                                             |
| <b>Business Rules</b>               | Must comply with encryption and zero-loss guarantee (NFR-5, NFR-3).                                                        |

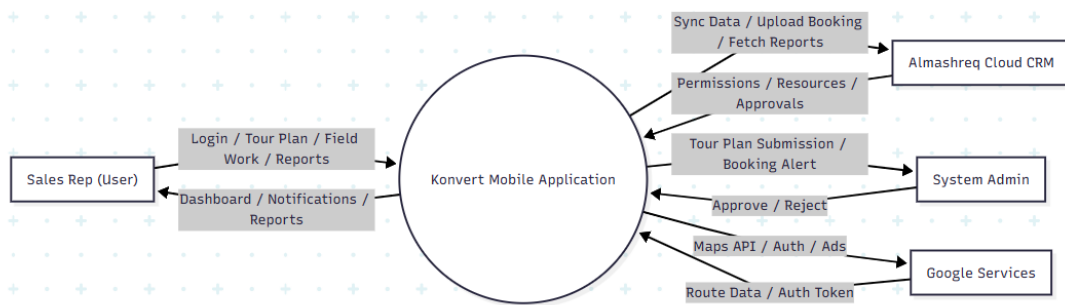
### 3.7

## Mock Screens

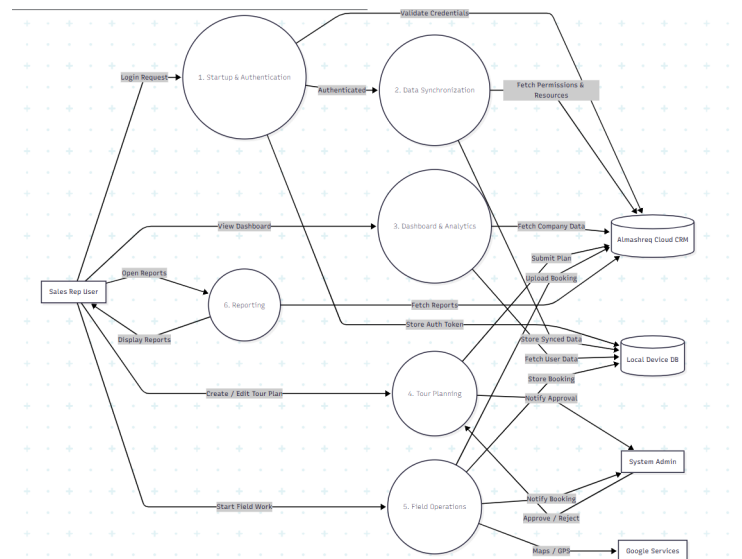
## Chapter 4

# Data Flow Diagrams

### 4.1 DFD Level 0



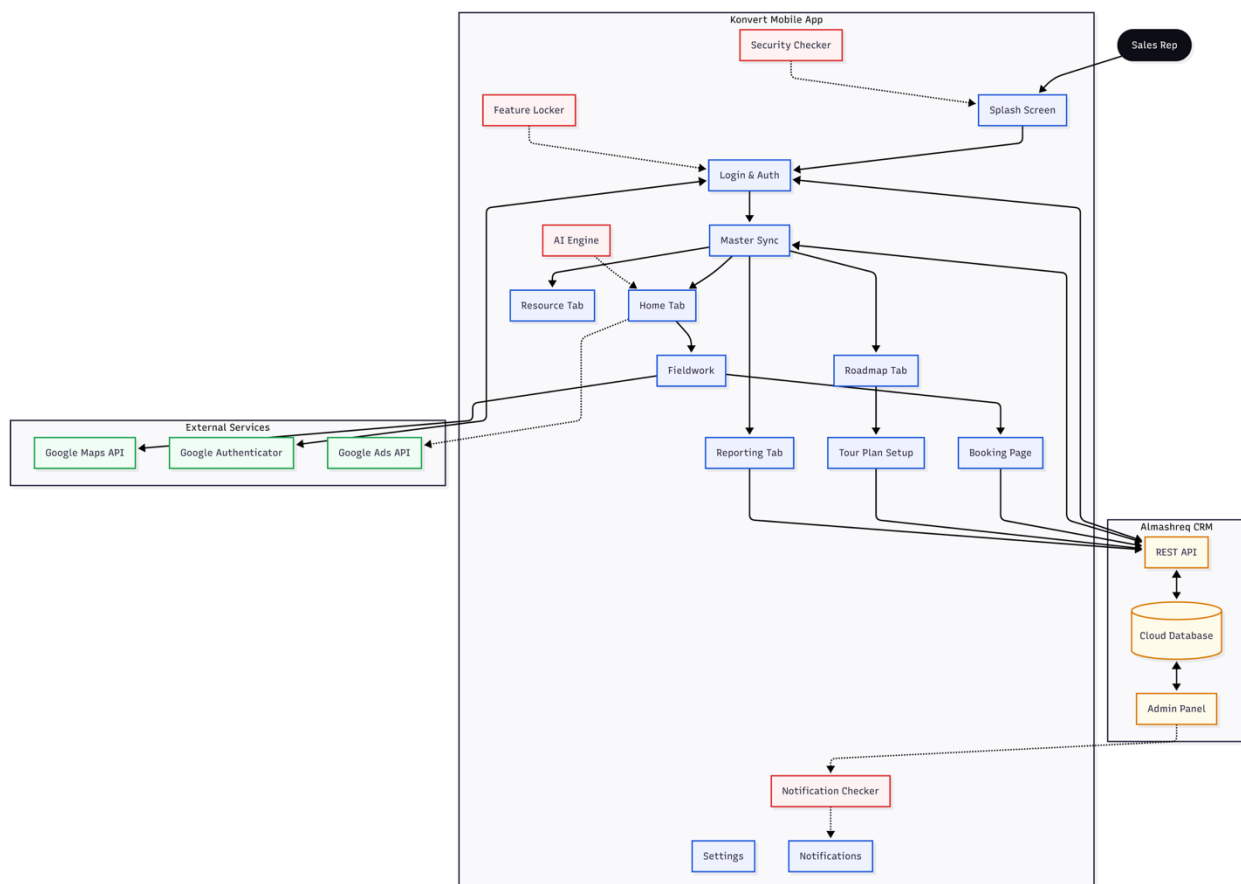
### 4.2 DFD Level 1



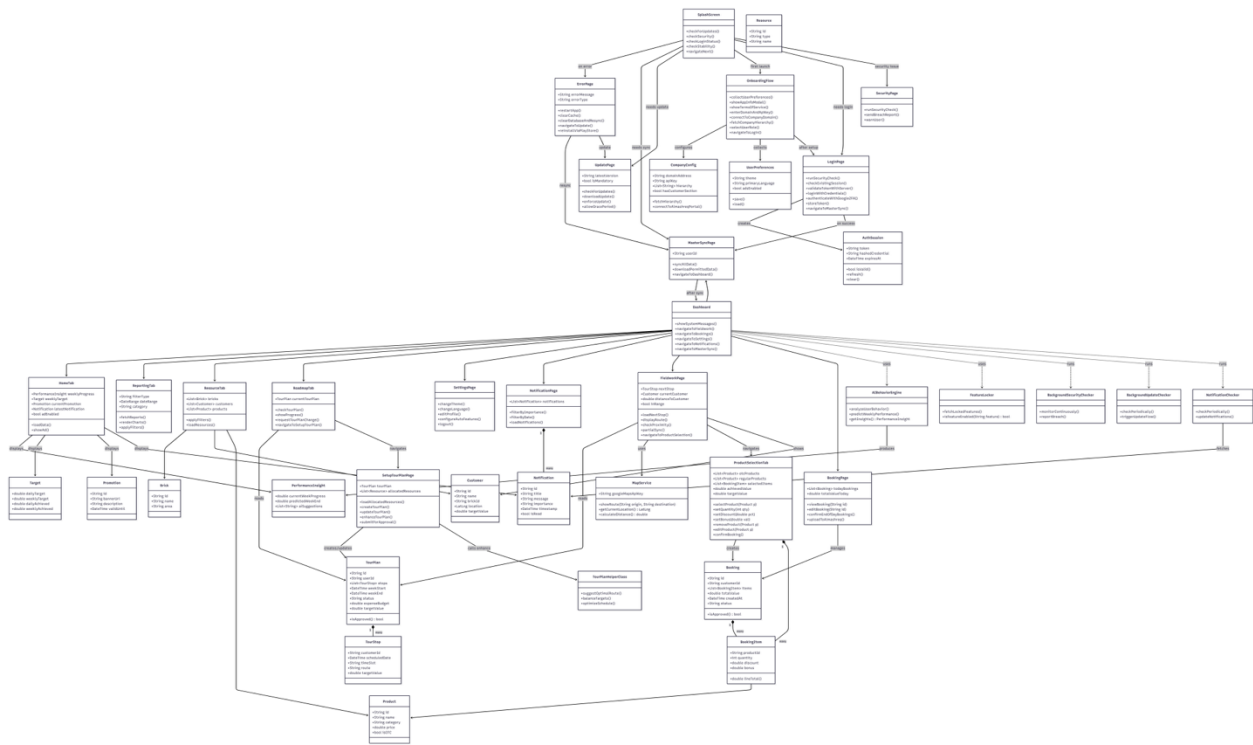


# System Design

## 5.1 System Architecture Diagram

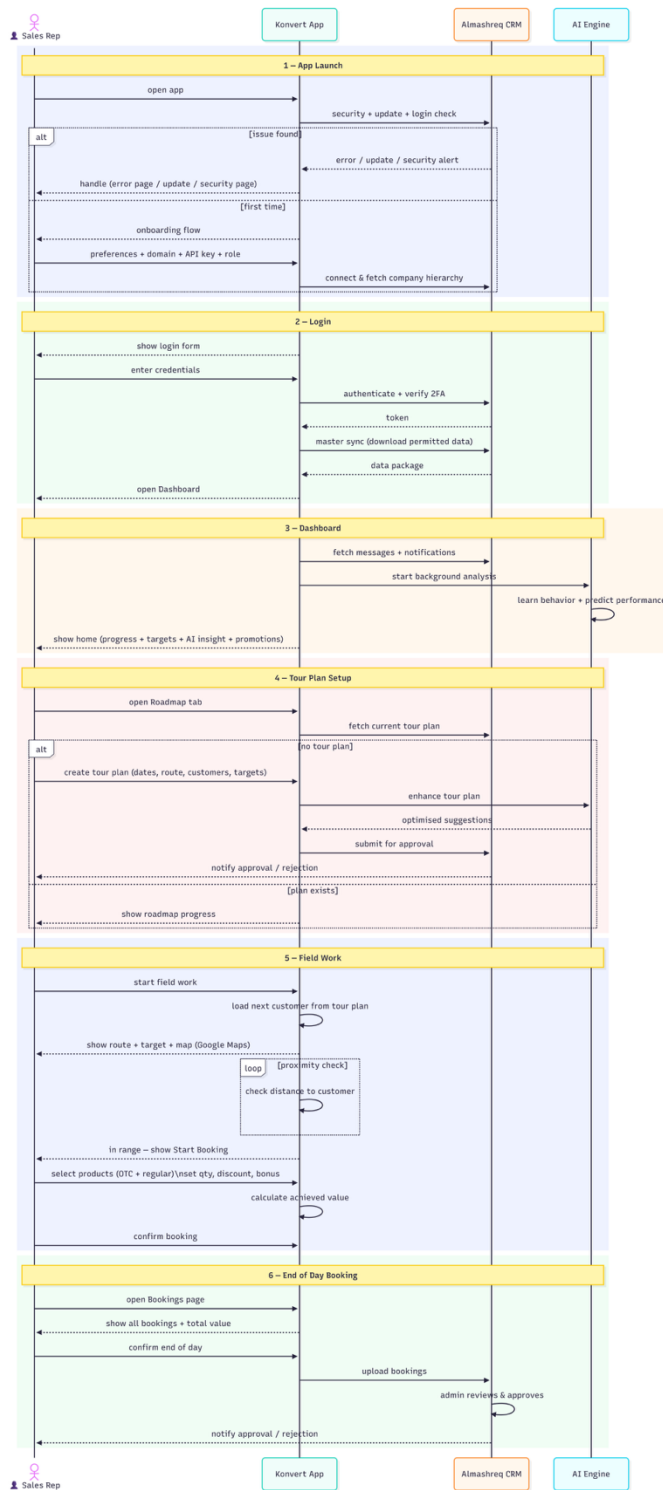


## 5.2 Class Diagram



## 5.3 ER Diagram

## 5.4 Sequence Diagram





## 5.5

### Data Dictionary

| Element Name   | Type     | Validation            | Mandatory | Remarks                                           |
|----------------|----------|-----------------------|-----------|---------------------------------------------------|
| UserID         | String   | Alphanumeric<br>UUID  | Yes       | Primary identifier synced from Almashreq.         |
| RoleTier       | Integer  | Range: 1 to 8         | Yes       | Defines the user's hierarchy level for RBAC.      |
| AuthToken      | String   | JWT Format            | Yes       | Token generated via Google Authenticator MFA.     |
| TourPlanID     | String   | Alphanumeric<br>UUID  | Yes       | Unique identifier for the assigned route plan.    |
| RouteDate      | Date     | YYYY-MM-DD            | Yes       | Scheduled date for the tour execution.            |
| CustomerID     | String   | Alphanumeric<br>UUID  | Yes       | Target entity for the field representative visit. |
| GeoCoordinates | GeoPoint | Valid Lat/Long        | Yes       | Captured at 5-minute intervals and check-ins.     |
| VisitStatus    | String   | Pending,<br>Completed | Yes       | Tracks the real-time status of a customer visit.  |
| ProductID      | String   | Alphanumeric<br>UUID  | Yes       | Unique identifier for OTC or regular products.    |
| OrderQuantity  | Integer  | Greater than 0        | Yes       | Number of units selected during booking.          |
| DiscountPct    | Float    | 0.0 to 100.0          | No        | Promotional discount applied during selection.    |

|            |          |                  |     |                                                  |
|------------|----------|------------------|-----|--------------------------------------------------|
| TotalValue | Float    | Greater than 0.0 | Yes | Final calculated amount of the daily booking.    |
| SyncState  | String   | Queued, Synced   | Yes | Indicator for offline-first conflict management. |
| CreatedAt  | DateTime | ISO 8601 format  | Yes | Exact timestamp of record creation/modification. |